

WASP-43b

R-0627

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-43_180118 · created 2026-07-25T14:12:11+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2458136.8678 +/- 0.0012 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.15 +/- 0.011
Transit depth (Rp/Rs)^2	2.24 +/- 0.33 [%]
Orbital Inclination (inc)	84.67 +/- 1.35
Airmass coefficient 1 (a1)	0.9759 +/- 0.0097
Airmass coefficient 2 (a2)	0.0153 +/- 0.0059
Scatter in the residuals of the lightcurve fit is	0.99 %
Best Comparison Star	#3 - [334, 61]
Optimal Aperture	3.94
Optimal Annulus	9.87
Transit Duration (day)	0.0566 +/- 0.0027

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.15 $\pm$ 0.011 vs 0.1594 (deviation 0.85 σ)
Duration fit vs published	0.0566 d vs 0.0512 d (deviation 1.98 σ)
Mid-time O–C	1.8 min
Depth SNR	13.6
Residual scatter	0.99 %

Light curve

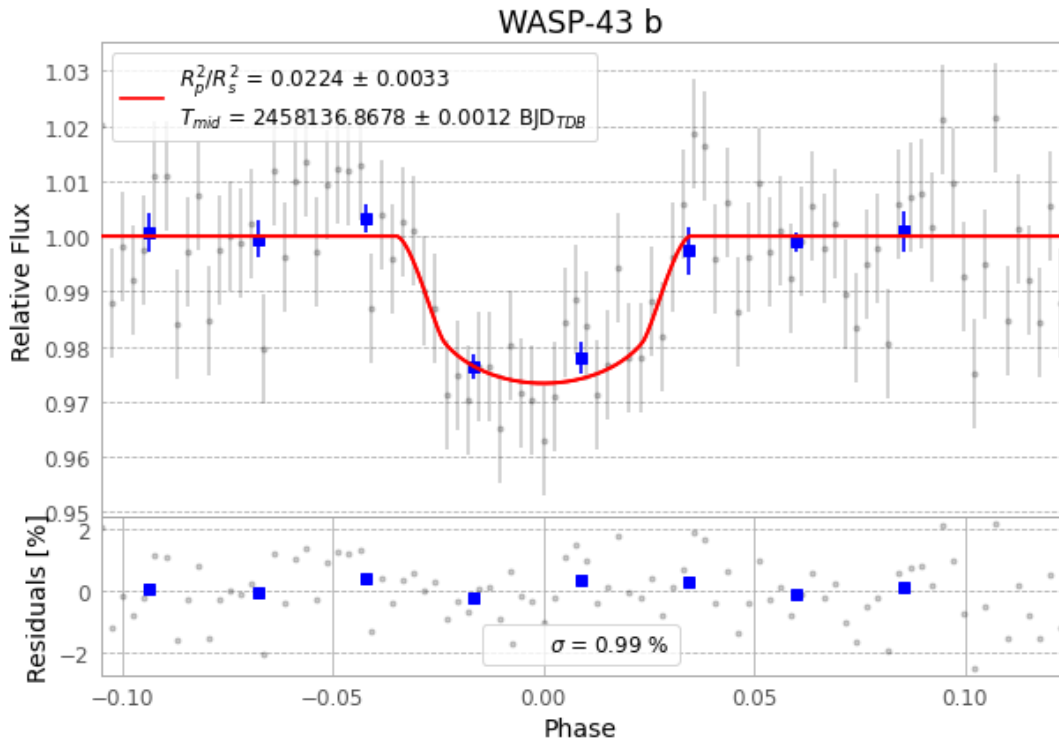


Figure 1 — FinalLightCurve_WASP-43 b_2018-01-17.png (SHA-256 5fa628cf0763906c...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [420, 231], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 546cc510f7b6031f...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit b365494

Data provenance

90 FITS frames (59 MB), WASP-43180118063912.FITS → WASP-43180118110612.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-180118.log (9ec1587670cc...), FinalLightCurve_WASP-43 b_2018-01-17.png (5fa628cf0763...), FinalLightCurve_WASP-43 b_2018-01-17.csv (4f83c1da0455...)

Compute resources

Wall time 115.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-25 14:12Z.